

📈 Experience

Research Assistant University of Rhode Island 📅 2021 – Present

- **Led end-to-end user research studies** investigating accessibility challenges in virtual reality (VR) systems for adults with intellectual and developmental disabilities (I/DD), including study design, recruitment, moderation, qualitative analysis, and reporting.
- Conducted **lab-based usability testing, interviews, and observational studies** with neurodiverse participants to understand usability barriers in immersive systems.
- Designed and implemented **research protocols, task scenarios, and evaluation metrics** to analyze how users perceive and interact with VR environments.
- Applied **thematic analysis** to synthesize qualitative data into actionable design insights used to guide product design decisions.
- Developed **user journey maps, accessibility guidelines, and design recommendations** that informed improvements in VR application interfaces.
- Collaborated with **external stakeholders and domain experts** to recruit participants and translate research findings into deployable applications.
- Co-developed and evaluated two mobile applications using participatory design:
Recognize, Report, Respond (R3) – abuse education mobile app for adults with I/DD App Store Amazon
Self-Care & Support (S2) – trauma-informed self-regulation support application App Store Google Play
- Produced research reports and presentations translating research findings into **actionable insights for design and development teams**.

Software Engineer Axiata Digital Labs/Dialog 📅 2017 – 2021

- Led backend development for **Genie**, a large-scale fintech mobile platform serving thousands of daily users. Collaborated with product and design teams to translate **user feedback and usage analytics** into improvements in service architecture and API performance.
- Improved system responsiveness and user experience by redesigning service communication patterns, reducing latency by **~40%**.
- Implemented containerized deployments using **Docker, Kubernetes, and Jenkins CI/CD pipelines**.
- Designed and implemented RESTful microservices for fintech platforms integrating **core banking systems (Finacle)**.
- Developed a public hotspot Wi-Fi authentication system and reduced authentication failures by **~35%** through system optimization.

🎓 Education

Rhode Island, USA University of Rhode Island 📅 Expected 2026

- Ph.D. in Computer Science (Human-Computer Interaction, Virtual Reality, Accessibility)

Sri Lanka Univ. of Sri Jayewardenepura 📅 2017

- B.Sc. in Computer Science (First Class Honors, GPA 3.85, **Gold Medalist**)

📄 Selected Publications

- **Kirupaharan, P.**, Ranalli, T.M., Venkatasubramanian, K. (2025). *Understanding the Accessibility of Single-User Virtual Reality Environments for Adults with Intellectual and Developmental Disabilities*. ACM DIS 2025.
- Lewis, B., **Kirupaharan, P.**, Venkatasubramanian, K. (2023). 'I... caught a person casing my house... and scared him off.': *The Use of Security-focused Smart Home Devices by PwD (Honorable Mention)*. CHI 2023.

💻 Research & Technical Skills

- **User Research Methods:** Usability Testing, Interviews, Surveys, Focus Groups, Field Studies, Qualitative Analysis, Thematic Analysis, Mixed Methods Research, Insight Synthesis, Story telling, Personas, User Journeys, Design Principles
- **Tools & Technologies:** Java, Python, Flutter, AWS, Docker, Kubernetes, LangChain, Unity